

CSE472 (Machine Learning Sessional)
Assignment 4: Principal component analysis (PCA)
and expectation-maximization(EM) algorithm

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#1905108

HOW TO RUN:

Ensure the Jupyter Notebook '1905108.ipynb' and Datasets Are in the Same Directory. Install pandas, numpy, scikit-learn, matplotlib if not already installed. Then run the notebook '1905108.ipynb'.

The implementations of PCA and EM can be found in the 'PCA' and 'EM' sections, respectively.

PLOTS:

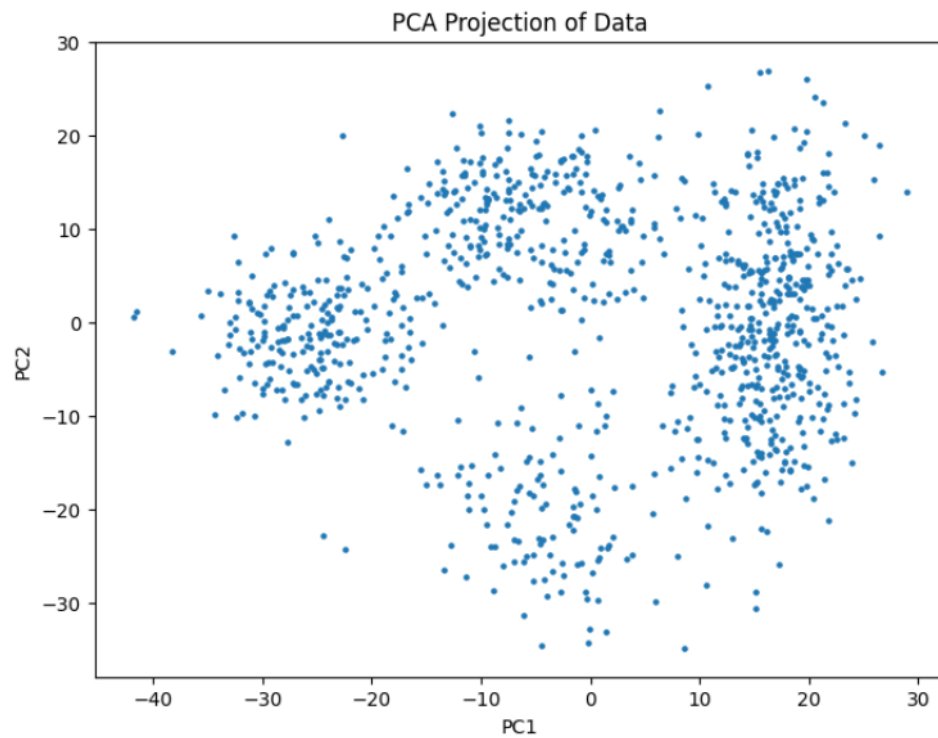


Fig.1: PCA projection of data

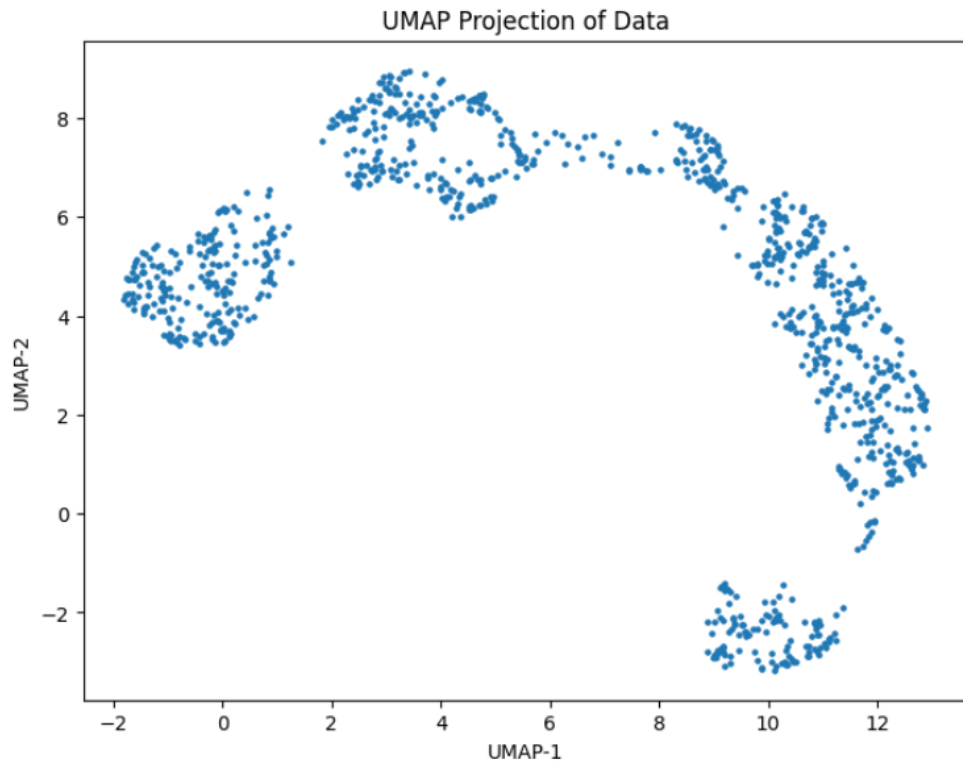


Fig.2: UMAP projection of data

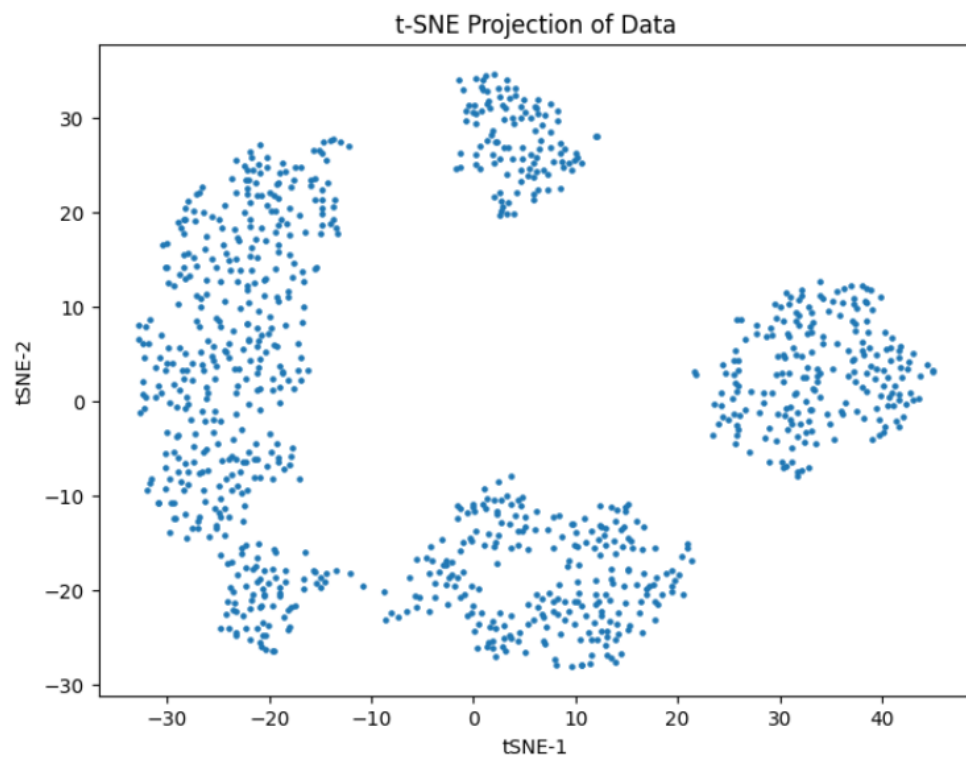


Fig.3: t-SNE projection of data

EXPECTATION-MAXIMIZATION (EM):

Converged at iteration: 179

Mean number of children (families with family planning): 1.78242202

Mean number of children (families without family planning): 4.91073091

Proportion of families with family planning: 0.35601692

Proportion of families without family planning: 0.64398308